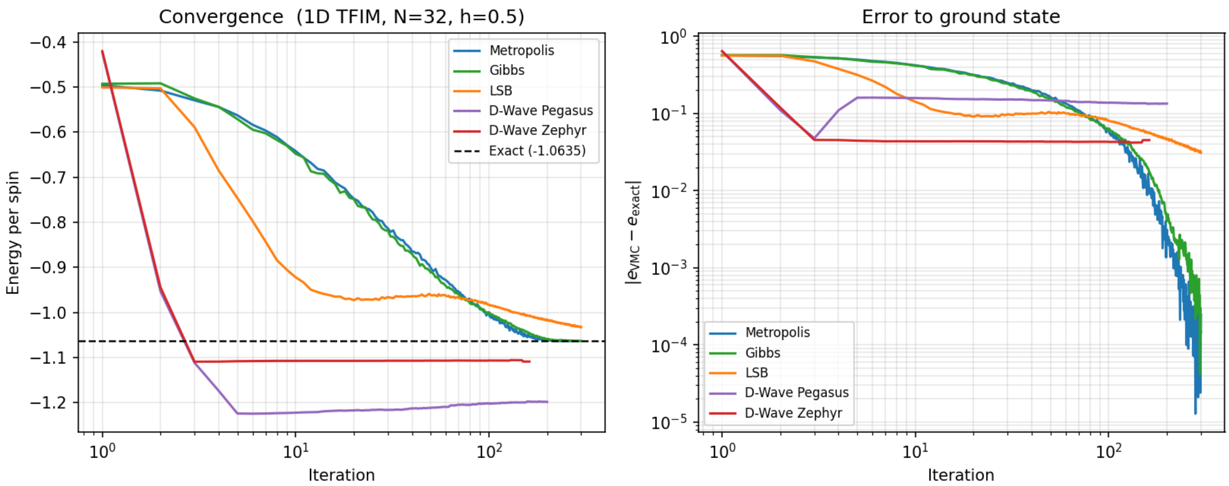
VMC/RBM Results Overview

1D TFIM and Long-Range 1D TFIM

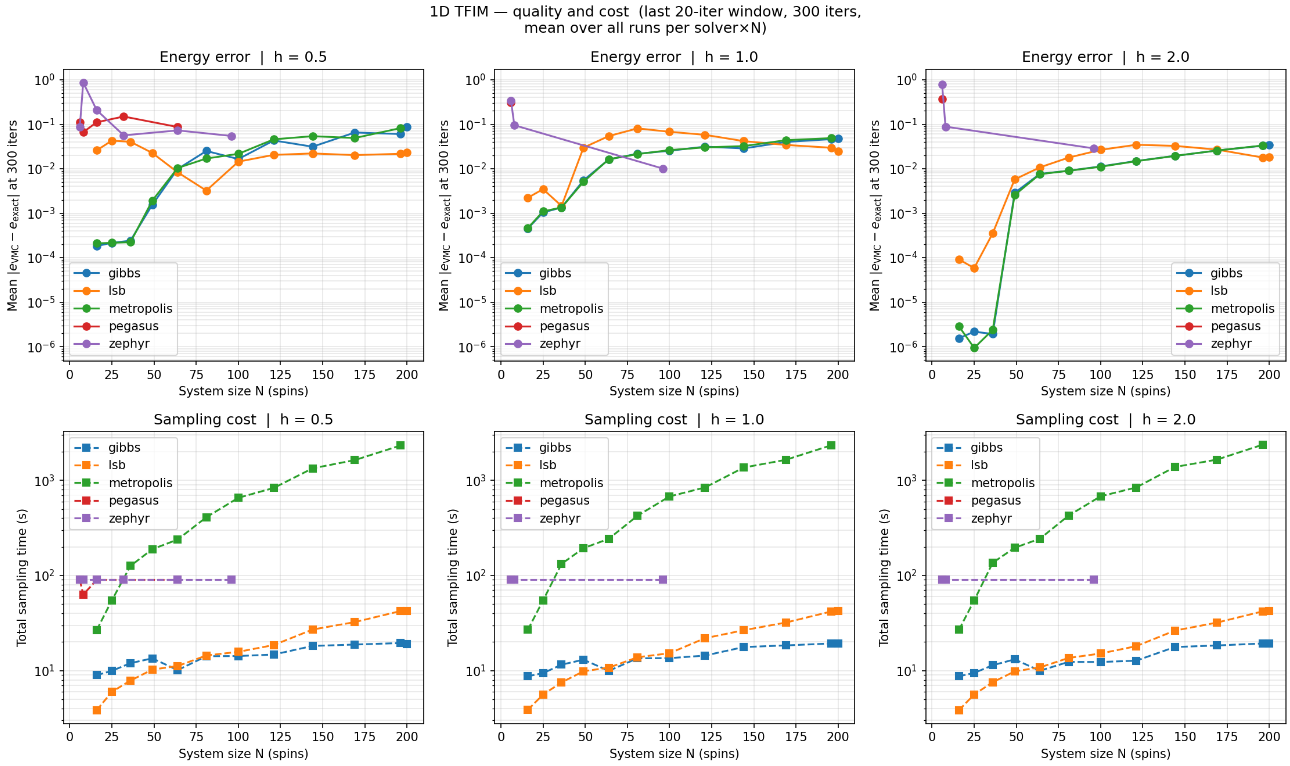
April 30, 2026

1D TFIM — Sampler Comparison (N = 32, h = 0.5, full RBM)



Energy per spin (left) and absolute error (right) for all sampler backends. Metropolis, Gibbs, and LSB converge to $\lesssim 10^{-4}$ per spin; D-Wave Pegasus and Zephyr plateau near 10^{-2}.

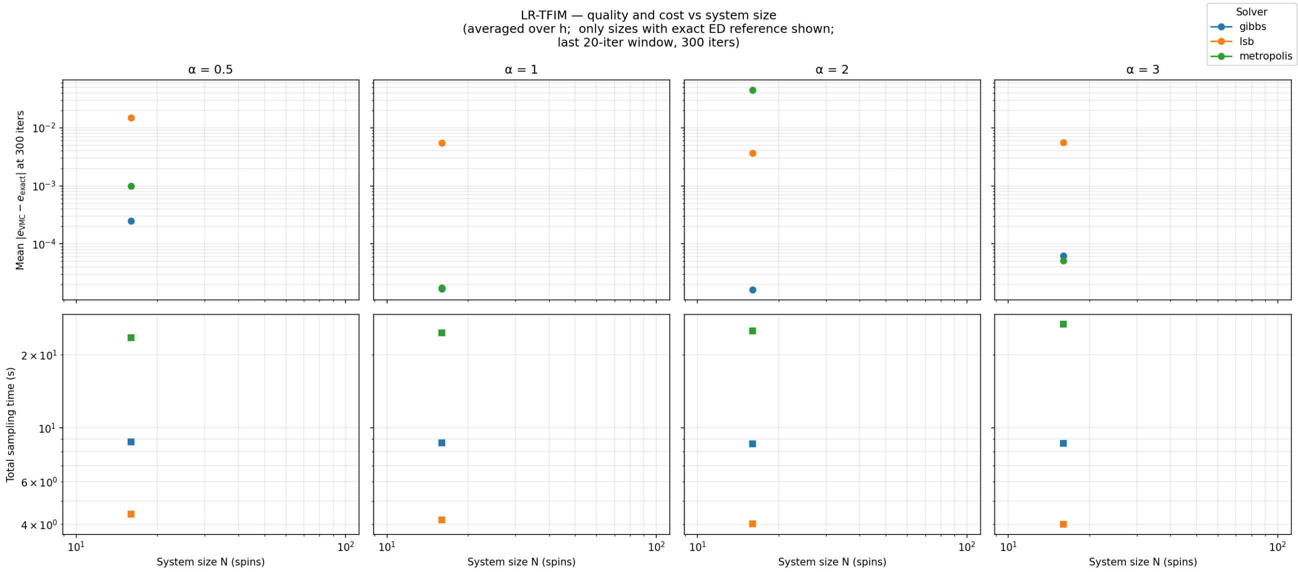
1D TFIM — Error and Cost vs System Size



Mean energy error per spin (top) and total sampling time (bottom) vs. N, for $h \in \{0.5, 1.0, 2.0\}$. Error measured in the last 20-iteration window of 300 iterations.

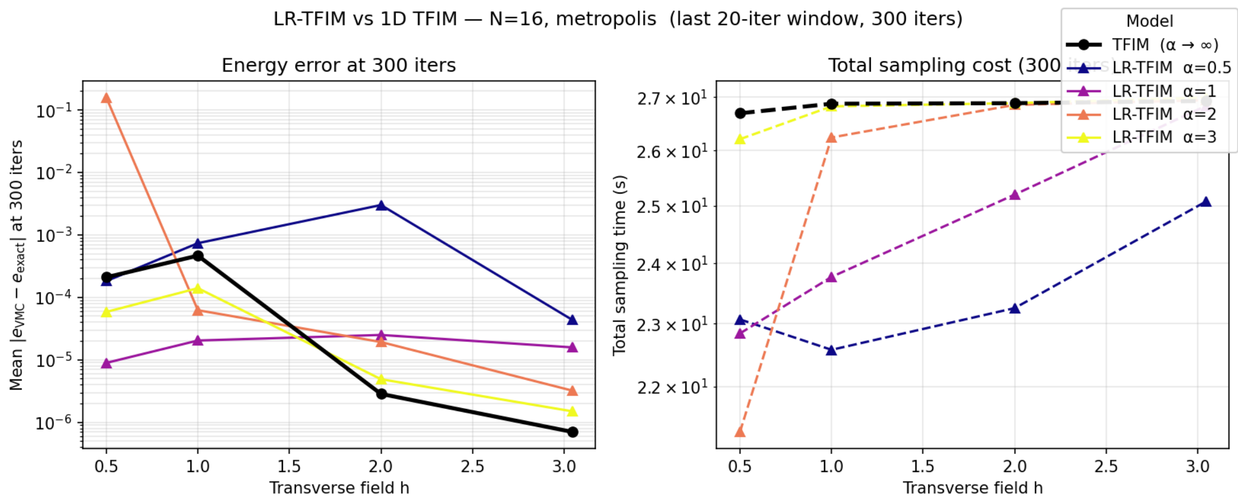
Long-Range 1D TFIM — Error and Cost vs System Size

$H = -J\sum_{i<j}\sigma_i^z\sigma_j^z/d(i,j)^\alpha - h\sum_i\sigma_i^x$, $\quad d(i,j) = \min(|i-j|, N-|i-j|)$. $\quad N \in \{16, 25, 36\}$; reference energy per spin from $N = 16$ ED for $N > 16$.



Same metrics as above, averaged over $h \in \{0.5, 1.0, 2.0, 3.044\}$, one column per α. Only $N = 16$ shown: exact ED reference only available there; larger N omitted because the per-spin ground state energy shifts with N (finite-size effect), making the N=16 reference an invalid baseline.

LR-TFIM vs. 1D TFIM — Direct Comparison at N = 16



Energy error (left) and total sampling time (right) vs. h, Metropolis sampler, last 20-iter window of 300 iterations. Black: nearest-neighbour TFIM ($\alpha \to \infty$). Coloured lines: LR-TFIM for $\alpha \in \{0.5, 1, 2, 3\}$.